The original limited augmented Zarankiewicz number in the $m \times 3$ case, and $z_L(14, 3) = 23$

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Abstract

The limited augmented Zarankiewicz number $z_L(m, n)$ of Qi, Cui and Xu is the largest value of $|E_1| + |E_2|$, where E_1 is a maximum C_4-free set of cells of the $m \times n$ grid and E_2 is a set of disjoint pairs of further cells, subject to the generalized C_4-cycle exclusions of their Definition 1. Values of $z_L(m, 3)$ were previously known only for $m \le 13$, the last seven of them from an exhaustive computer search by Qi, Löfberg and Chen; the value $z_L(14, 3)$ and all larger values were open. We prove that

$$z_L(m, 3) = \left\lfloor \frac{3m + 4}{2} \right\rfloor = m + 3 + \left\lfloor \frac{m - 2}{2} \right\rfloor \qquad \text{for every } m \ge 3.$$

In particular $z_L(14, 3) = 23$. The proof needs no computer. Its key step is a short lemma: in every admissible limited configuration with three columns, at most one row is fully occupied. Counting the cells occupied by 2-edges then gives the upper bound. An explicit family of configurations attains it. The formula agrees with all eleven published values for $3 \le m \le 13$. It also shows that the gap between the weak and the original limited numbers, which equals 2 or 3 for $7 \le m \le 13$, grows like $m/6$. We check the result independently by an exhaustive branch-and-bound search and by a separate SAT encoding at $m = 14$, over every extremal base graph.

1 Introduction

Let $\mathrm{BSR}(m, n)$ denote the maximum sum-of-squares rank of an $m \times n$ SOS biquadratic form. Qi, Cui and Xu [1] introduced *augmented bipartite graphs*. Such a graph has ordinary edges (1-edges), which encode squares $x_i^2 y_j^2$, and *2-edges*, which encode squares of binomials $(x_i y_j + x_k y_l)^2$. They found combinatorial conditions, the absence of a *generalized C_4-cycle*, that make the associated "doubly simple" form irreducible. This gives the chain $\mathrm{BSR}(m, n) \ge z_A(m, n) \ge z_L(m, n) \ge z(m, n)$, where $z(m, n)$ is the classical Zarankiewicz number. The *limited* number z_L requires the 1-edges to form a maximum C_4-free graph.

For three columns the classical value is $z(m, 3) = m + 3$ for $m \ge 3$ (Čulík [8]; see also [9]). The known original limited values were

$$z_L(3, 3) = 6,\ z_L(4, 3) = 8\ [1], \quad z_L(5, 3) = 9\ [2], \quad z_L(6, 3) = 11\ [3, 5],$$

and, by exhaustive computation, Qi, Löfberg and Chen [5] extended the list to

$$z_L(7, 3), \ldots, z_L(13, 3) = 12, 14, 15, 17, 18, 20, 21.$$

They listed the determination of $z_L(m, 3)$ for all m as an open problem [5, §9, Problem 1]. They also asked whether the gap $z_{wL}(m, 3) - z_L(m, 3)$, which they observed to be 2 or 3 for $7 \le m \le 13$, persists. Here z_{wL} is the *weak* limited number of [4], which they determined for all m. The weak and the original numbers are different quantities. This note concerns only the original (strong) one.

Theorem 1 (Main theorem)**.** *For every integer $m \geq 3$,*

$$z_L(m,3) = m + 3 + \left\lfloor \frac{m-2}{2} \right\rfloor = \left\lfloor \frac{3m+4}{2} \right\rfloor .$$

In particular $z_L(14,3) = 17 + 6 = 23$.

The proof (Sections 4 and 5) is short and analytic. The upper bound rests on Lemma 6: at most one row of an admissible configuration can be completely occupied. Section 7 describes two independent exhaustive computations at $m = 14$, which we ran as an independent check and which are not used in the proof. Section 6 discusses the consequences for the weak/strong gap.

2 Definitions

We use the language of the grid $[m] \times [n]$. Its elements (i, j) are *cells*, with i the row and j the column. A bipartite graph between rows and columns is the same as a set of cells. A *2-edge* is an unordered pair $\{(i,j),(k,l)\}$ of distinct cells, written $(i, j; k, l)$; its two cells are its *halves*. It is *nondegenerate* if $i \neq k$ and $j \neq l$, and *degenerate* if the two halves share a row or a column.

Definition 2 (Qi–Cui–Xu [1, Definition 1])**.** Let E_1 be a C_4-free set of cells and E_2 a set of 2-edges satisfying the simplicity condition

(S) no 2-edge shares a cell with a 1-edge or with another 2-edge.

A cell is *occupied* if it lies in E_1 or is a half of some 2-edge; let $\mathcal{O}$ be the set of occupied cells. The configuration contains a *generalized C_4-cycle* if one of the following holds:

(C1) four cells of E_1 form a rectangle (a classical C_4);

(C2) some nondegenerate 2-edge $(i, j; k, l)$ has both opposite cells (i, l) and (k, j) occupied;

(C3) there are a 2-edge $(i, j; p, q)$ of any type and a cell (k, l) with $k \notin \{i, p\}$ and $l \notin \{j, q\}$ such that the cells $(k,l),(k,j),(k,q),(i,l),(p,l)$ are all occupied; if the 2-edge is nondegenerate these five cells are required to be pairwise distinct.

The configuration is *limited* if $|E_1| = z(m, n)$. It is *admissible* if it satisfies (S) and contains no generalized C_4-cycle. Then $z_L(m, n)$ is the maximum of $|E_1| + |E_2|$ over admissible limited configurations.

Remark 3 (Fidelity of the transcription)**.** Definition 2 follows [1, Definition 1] literally. Two points deserve comment.

(a) For a nondegenerate 2-edge the distinctness clause in (C3) holds automatically. The conditions $i \neq p$, $j \neq q$, $k \notin \{i, p\}$ and $l \notin \{j, q\}$ already make the five cells distinct. For a degenerate 2-edge two of the five cells coincide, and (C3) is applied to the resulting four cells. This is the reading of [1]: the text says "a 2-edge ... (of any type)", and Remark 1 there says that (C3) "applies uniformly to all 2-edge types".

(b) The alternative reading, in which (C3) is never applied to degenerate 2-edges, contradicts the published values. Our search program under that reading returns $9, 11, 13, 15, 17, 19$ for $m = 4, \ldots, 9$. The published values are $8, 9, 11, 12, 14, 15$; in particular, $z_L(4, 3) = 8$ is a theorem of [1].

With Definition 2 as stated, our independent program reproduces all eleven published values $z_L(m, 3)$, $3 \leq m \leq 13$ (Section 7). The weak conditions of [4] (dependency digraphs, complementary 2-cycles) play no role in this note.

Every condition in Definition 2 is invariant under permutations of the rows and under permutations of the columns. Moreover, (C2) and (C3) are *monotone*: if a 2-edge e has a (C2) or (C3) configuration with respect to a set $\mathcal{O}$ of occupied cells, it still has one with respect to every larger set. Consequently, deleting 2-edges from an admissible configuration keeps it admissible.

3 The extremal base graphs

From now on $n = 3$ and $m \geq 3$. The following classification is classical (see [8, 9], and [5, Theorem 3]). We include the proof because the rest of the argument uses its exact form.

Theorem 4. *$z(m, 3) = m + 3$. A C_4-free set $E_1 \subseteq [m] \times [3]$ has $m + 3$ cells if and only if exactly three rows contain two cells of E_1, with column supports $\{2, 3\}$, $\{1, 3\}$ and $\{1, 2\}$, and every other row contains exactly one cell of E_1.*

Proof. Let d_r be the number of cells of E_1 in row r. Two rows containing the same pair of columns would form a C_4. Hence $\sum_r \binom{d_r}{2} \leq \binom{3}{2} = 3$. For every integer $d \geq 0$ we have $d \leq 1 + \binom{d}{2}$, with equality if and only if $d \in \{1, 2\}$. Summing, $|E_1| = \sum_r d_r \leq m + \sum_r \binom{d_r}{2} \leq m + 3$. Equality forces every $d_r \in \{1, 2\}$ and $\sum_r \binom{d_r}{2} = 3$. So exactly three rows have degree two, and their supports are the three distinct pairs. Conversely, such a set is C_4-free: no two rows share a pair of columns, since only the three special rows contain pairs, and their pairs are distinct. □

Fix an extremal base E_1. For $t \in \{1, 2, 3\}$ let S_t be the *special* row whose support misses column t. Its unique free cell is $Q_t = (S_t, t)$. Each of the other $m - 3$ rows is a *singleton* row. It contains one 1-edge, in a column called its *colour*, and two free cells. We record two facts used repeatedly:

(F1) every row contains at least one 1-edge and at least one free cell;

(F2) for any two distinct columns a, b, the special row whose support is $\{a, b\}$ has 1-edges in both columns a and b.

In particular every limited configuration has $2m - 3$ free cells, and an admissible configuration with $s = |E_2|$ occupies exactly $2s$ of them.

A row is *full* if all three of its cells are occupied.

4 The upper bound

We say that a 2-edge e *triggers* (C3) if a cell (k, l) as in (C3) exists for e, so that admissibility means that no 2-edge triggers (C3). With three columns, condition (C3) only constrains the two rows of the 2-edge together with one other row.

Lemma 5 ((C3) for three columns)**.** *Let e be a 2-edge of a configuration on $[m] \times [3]$.*

(a) If $e = (i, j; p, q)$ is nondegenerate and l is the column not in $\{j, q\}$, then e triggers (C3) if and only if there is a full row $k \notin \{i, p\}$ while $(i, l), (p, l) \in \mathcal{O}$.

(b) If $e = (i, j; i, q)$ lies in a single row i, then e triggers (C3) if and only if row i is full and some other row is full.

(c) If $e = (i, j; p, j)$ lies in a single column j, then e triggers (C3) if and only if, for some column $l \neq j$ and some row $k \notin \{i, p\}$, the four cells $(k, l), (k, j), (i, l), (p, l)$ are occupied.

Proof. (a) Here $l \notin \{j, q\}$ forces l to be the third column. Then $(k, l), (k, j), (k, q)$ is the whole row k, so the (C3) configuration says exactly that row k is full and $(i, l), (p, l) \in \mathcal{O}$. (b) With

$i = p$ the five cells reduce to row k (all three cells) and the cell (i, l). Since (i, j) and (i, q) are halves, $(i, l) \in \mathcal{O}$ means that row i is full. (c) With $j = q$ the five cells reduce to the four listed cells. □

Lemma 6 (Unique full row)**.** *In every admissible limited configuration on* $[m] \times [3]$*,* $m \geq 3$*, at most one row is full.*

Proof. Let r be a full row. By (F1) it contains a free cell. Since r is full, that cell is occupied, so it is a half $u = (r, a)$ of some 2-edge e. Let $v = (p, x)$ be the other half of e. Let b, c be the two columns other than a. Since r is full, $(r, b), (r, c) \in \mathcal{O}$. We show that no row other than r is full.

Case 1: $p = r$. Then e lies in row r, which is full. By Lemma 5(b), admissibility forbids any other full row.

Case 2: $p \neq r$ *and* $x = a$. This case is impossible. Here e lies in column a. By (F2), the special row K with support $\{a, b\}$ has $(K, a), (K, b) \in E_1$. Note $K \neq r$: either r is a singleton row, or r is the special row whose free cell is in column a, namely $S_a \neq K$. Also $K \neq p$, because (p, a) is free while (K, a) is a 1-edge. Apply Lemma 5(c) with $l = b$ and $k = K$. Three of the four cells, (K, b), (K, a) and (r, b), are occupied, so admissibility forces $(p, b) \notin \mathcal{O}$. Arguing in the same way with the special row of support $\{a, c\}$ gives $(p, c) \notin \mathcal{O}$. But (p, a) is free, and by (F1) row p has a 1-edge, which must then lie in column b or c. This is a contradiction.

Case 3: $p \neq r$ *and* $x \neq a$. Then e is nondegenerate. Its opposite cells are (r, x) and (p, a). Since $x \in \{b, c\}$, $(r, x) \in \mathcal{O}$, so (C2) forces $(p, a) \notin \mathcal{O}$. In particular row p is not full and (p, a) is not a 1-edge. Let l be the column outside $\{a, x\}$. Row p has a 1-edge by (F1), and it is not in column x (where (p, x) is free) or in column a. Hence $(p, l) \in E_1 \subseteq \mathcal{O}$. Also $(r, l) \in \mathcal{O}$ because $l \in \{b, c\}$. By Lemma 5(a) there is no full row outside $\{r, p\}$. Since p is not full, r is the only full row. □

Theorem 7 (Upper bound)**.** *Every admissible limited configuration on* $[m] \times [3]$*,* $m \geq 3$*, has* $|E_2| \leq \lfloor (m-2)/2 \rfloor$*. Hence* $z_L(m, 3) \leq m + 3 + \lfloor (m-2)/2 \rfloor$*.*

Proof. For a row ρ let $f(\rho)$ be its number of free cells, so $f = 1$ for the three special rows and $f = 2$ for the $m - 3$ singleton rows. Let $h(\rho)$ be the number of its free cells that are occupied. By (S), each 2-edge occupies two distinct free cells and no cell is shared, so $2|E_2| = \sum_\rho h(\rho)$. Every row has $h(\rho) \leq f(\rho) - 1$ unless it is full, in which case $h(\rho) = f(\rho)$. By Lemma 6,

$$2|E_2| \leq \sum_\rho (f(\rho) - 1) + \#\{\text{full rows}\} \leq (m-3) \cdot 1 + 3 \cdot 0 + 1 = m - 2.$$

Since $|E_2|$ is an integer, $|E_2| \leq \lfloor (m-2)/2 \rfloor$, and $|E_1| + |E_2| \leq m + 3 + \lfloor (m-2)/2 \rfloor$ by Theorem 4. □

The proof also shows what equality requires. When m is even, an optimal configuration has exactly one full row, every singleton row other than it contains exactly one half, and no non-full special row contains a half.

5 The matching construction

Write $N = m - 3 = 2h + \varepsilon$ with $\varepsilon \in \{0, 1\}$. Rows $1, 2, 3$ are the special rows S_1, S_2, S_3, with $E_1 \cap (\text{rows } 1, 2, 3) = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$. The remaining rows are:

- for $t = 1, \ldots, h$, a row R_t with 1-edge $(R_t, 3)$ and a row P_t with 1-edge $(P_t, 1)$, joined by the 2-edge $e_t = (R_t, 2;\ P_t, 3)$;
- if $\varepsilon = 1$, one more row U with 1-edge $(U, 1)$ and the 2-edge $g = (S_3, 3;\ U, 2)$, which uses the free cell Q_3.

Thus $|E_1| = m + 3$ and $|E_2| = h + \varepsilon = \lceil N/2 \rceil = \lfloor (m-2)/2 \rfloor$. Table 1 shows the case $m = 14$.

Proposition 8. *The configuration above is admissible and limited. Hence* $z_L(m, 3) \geq m + 3 + \lfloor (m-2)/2 \rfloor$ *for all* $m \geq 3$.

Proof. E_1 has the form of Theorem 4, so it is C_4-free with $|E_1| = z(m, 3)$, and (C1) fails. The halves listed are distinct free cells, so (S) holds. Each singleton row contains exactly one half, and the free cells Q_1, Q_2 are empty. So the only full row is S_3, and only when $\varepsilon = 1$.

The edge e_t. Its opposite cells are $(R_t, 3)$ and $(P_t, 2)$. The free cells of P_t are $(P_t, 2)$ and $(P_t, 3)$, and only $(P_t, 3)$ is a half, so $(P_t, 2) \notin \mathcal{O}$ and (C2) fails. The third column of e_t is $l = 1$. The free cells of R_t are $(R_t, 1)$ and $(R_t, 2)$, and only $(R_t, 2)$ is a half, so $(R_t, 1) \notin \mathcal{O}$. By Lemma 5(a), e_t does not trigger (C3).

The edge g (if $\varepsilon = 1$). Its opposite cells are $(S_3, 2)$ and $(U, 3)$, and $(U, 3) \notin \mathcal{O}$, so (C2) fails. The third column is $l = 1$, and $(S_3, 1), (U, 1) \in E_1$. By Lemma 5(a), g would trigger (C3) only if some row outside $\{S_3, U\}$ were full, and there is none.

Thus no generalized C_4-cycle exists. □

row	role	col. 1	col. 2	col. 3	
1	S_1	·	■	■	
2	S_2	■	·	■	
3	S_3	■	■	g	full row
4,6,8,10,12	$R_1, \ldots, R_5$	·	e_t	■	
5,7,9,11,13	$P_1, \ldots, P_5$	■	·	e_t	
14	U	■	g	·	

Table 1: An optimal configuration for $m = 14$: ■ is a 1-edge, · is an empty cell, and a label marks a half of the named 2-edge. The 2-edges are $e_t = (2t+2, 2;\ 2t+3, 3)$ for $t = 1, \ldots, 5$ and $g = (3, 3; 14, 2)$. Here $|E_1| = 17$, $|E_2| = 6$ and the total is 23. The base type is $(a, b, c) = (6, 0, 5)$ singleton rows of colours $1, 2, 3$, and the 25 free cells contain 12 halves and 13 empty cells.

Proof of Theorem 1. Combine Theorem 7 and Proposition 8. For $m = 14$ this gives $z_L(14, 3) = 17 + \lfloor 12/2 \rfloor = 23$. □

Remark 9. The upper bound holds for every labelled extremal base, so the proof needs neither a classification up to symmetry nor any symmetry breaking. The formula also gives every published value: $\lfloor (3m+4)/2 \rfloor = 6, 8, 9, 11, 12, 14, 15, 17, 18, 20, 21$ for $m = 3, \ldots, 13$. For $m = 2$ the base structure is different, since $z(2, 3) = 4$, and the formula does not apply.

6 Consequences

The weak/strong gap. Qi, Löfberg and Chen [5] proved $z_{wL}(m, 3) = m + 3 + \lceil m/2 \rceil + 1$ for $9 \leq m \leq 15$ and $z_{wL}(m, 3) = m + 3 + \lfloor (2m-4)/3 \rfloor$ for $m \geq 16$. Together with Theorem 1 this gives

$$z_{wL}(m, 3) - z_L(m, 3) = \begin{cases} \lceil m/2 \rceil + 1 - \lfloor (m-2)/2 \rfloor \in \{2, 3\}, & 9 \leq m \leq 15, \\ \lfloor (2m-4)/3 \rfloor - \lfloor (m-2)/2 \rfloor, & m \geq 16. \end{cases}$$

For example, $z_{wL}(14, 3) - z_L(14, 3) = 25 - 23 = 2$. For $m \geq 16$ the gap is unbounded and grows like $m/6$. The bounded gap of 2 or 3 seen for $7 \leq m \leq 13$ therefore does not persist, which answers the question in [5, §9, Problem 1]. The densities are $\lim_{m\to\infty} z_L(m, 3)/m = 3/2$ and $\lim_{m\to\infty} z_{wL}(m, 3)/m = 5/3$.

SOS rank. Since $z_{wL}(m,3) \geq z_L(m,3)$, Theorem 1 gives no new lower bound for $\mathrm{BSR}(m,3)$. Its value lies in settling the original framework of [1] completely for three columns.

Where the strong framework loses. Lemma 6 explains the gap structurally. In the weak framework of [4, 5], condition (W3) involves only 1-edges and pairs of vertex-disjoint edges. Many full rows can then coexist, for example via complementary pairs. In the original condition (C3), any occupied cell, half or not, can serve as the cell (k,l). With three columns, a single full row then blocks every 2-edge whose two rows are both occupied in its third column.

7 Independent computational verification

The proof above is complete without computation. As an independent check of both the theorem and our reading of Definition 2, we wrote three programs. Their source and outputs accompany this note (see the reproducibility section).

Literal checker. This routine implements Definition 2 verbatim. It checks (S), (C1), (C2) and (C3) by iterating over all 2-edges and all cells (k,l), and uses no lemma of this note. It validates every configuration reported below, and it validates the construction of Section 5 for all $3 \leq m \leq 200$.

Method A: exhaustive branch and bound. For each base type (a,b,c) with $a \geq b \geq c$ and $a+b+c = m-3$, the search places 2-edges among the free cells. It processes the free cells in a fixed order, and each cell is either left empty or paired with a later undecided free cell. The pruning uses only monotonicity: a branch is cut as soon as a fully occupied witness set (from (C2) or (C3)) appears. The search also discards branches that cannot beat the best value already found, and the bound it uses for this is a valid upper bound. By Theorem 4 and the permutation invariance of Definition 2, the base types represent every extremal base. Within a base the search does no symmetry breaking. It reproduces all published values for $3 \leq m \leq 13$ exactly. For $m = 14$ it gives the per-base maxima in Table 2, with overall maximum 6.

Method B: SAT encoding. This program shares no code with Method A. It re-derives the base types and introduces a Boolean variable for every unordered pair of free cells, without pre-filtering. It encodes (S) by at-most-one constraints and (C2) and (C3) as implications on occupancy literals, and asks the SMT solver z3 whether $|E_2| \geq s$ is satisfiable. At $m = 14$ it finds models with $s = 6$ on exactly the base types where Method A finds 6. It reports $s \geq 6$ unsatisfiable on the others, and $s \geq 7$ unsatisfiable on all sixteen base types (file `certificates/z3_m14_s7.json`).

Exhaustive check of Lemma 6. For $3 \leq m \leq 7$ we enumerated every admissible limited configuration on every *labelled* extremal base, using only the literal checker. There are 6, 288, 5220, 125280 and 2262330 such configurations for $m = 3,4,5,6,7$, and none has two full rows. For $3 \leq m \leq 8$, brute force over all cell subsets confirms Theorem 4.

Reproducibility

The accompanying package contains these Python sources (paths relative to the package root) and their JSON outputs in `certificates/`:

- `src/zl_core.py`: the literal checker of Definition 2;
- `src/search_dfs.py`: Method A;

$\max \lvert E_2 \rvert$	base types (a, b, c)
5	$(11,0,0)$, $(10,1,0)$, $(9,2,0)$, $(9,1,1)$, $(8,3,0)$, $(8,2,1)$
6	$(7,4,0)$, $(7,3,1)$, $(7,2,2)$, $(6,5,0)$, $(6,4,1)$, $(6,3,2)$, $(5,5,1)$, $(5,4,2)$, $(5,3,3)$, $(4,4,3)$

Table 2: Maximum number of 2-edges on each of the sixteen extremal base types for $m = 14$ (Method A, confirmed by Method B). The type (a, b, c) lists the numbers of singleton rows of the three colours, up to column permutation. The overall maximum 6 agrees with Theorem 7. The table is computational and is not used in the proof.

- `src/verify_z3.py`: Method B;
- `src/construction.py`: the family of Section 5;
- `src/verify_lemmas.py`: the exhaustive small-m checks;
- `src/make_certificates.py`: regenerates the certificates;
- `src/check_package.py`: reruns the fast checks.

The programs need Python 3.9 or later. Method B also needs the `z3-solver` package.

Scope, novelty and authorship

The result concerns the original limited number of [1, Definition 1] for $n = 3$, and covers all $m \geq 3$. We checked the primary sources and the later related preprints [5, 4, 3, 2, 7, 6] on arXiv (accessed 29 October 2026). Before this note, $z_L(m, 3)$ was known only for $m \leq 13$, and we found no formula for general m. This note was produced by Claude Opus 5.5, an AI system developed by Anthropic, which carried out the derivation, the programs and the writing. The verification consists of the internal checks described above. No external peer review and no proof-assistant formalization is claimed.

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